DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

"Data Science Intern"

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

DEV KUMAR KHETAN - 1DS20AI023

Conducted at

Neuberg Anand Academy of Laboratory Medicine

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangalore-78

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

CERTIFICATE

This is to certify that the Internship titled **"Data Science Intern"** carried out by **Mr DEV KUMAR KHETAN [1DS20AI023]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (20AI8ICINT)

Suma.S.N

Signature of Reporting Head	**Signature of Internship Coordinator**	**Signature of HoD**
Suma S N	**Prof. Kavya D N**	**Dr. Vindhya P Malagi**
Data Scientist	Dept. of AI & ML	Dept. of AI & ML
Anand Diagnostic Laboratory	DSCE	DSCE

External Viva:

Name of the Examiner	Signature with Date
1) ________________	________________
2) ________________	________________

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning

D E C L A R A T I O N

I, **DEV KUMAR KHETAN,** final year student of Information Science and Engineering, Dayananda Sagar College Of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **Neuberg Anand Academy of Laboratory Medicine**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:22/04/2024 USN : 1DS20AI023

Place: Bengaluru Name : Dev Kumar Khetan

OFFER LETTER

Internship Documentation Process Inbox ×

Deepak Krishna naik <deepakkn@neubergdiagnostics.com> Wed, Mar 29, 2023, 4:46 PM

to nishank.satish, me, sahanarajgowda1801, 20beam012, mgakhil03, maazkarim02, pbs.dsce2001, ashifalam470, Rashmi-aiml, renukaedvi-aiml, Keerti, Ms, Divakar, Vindhya, Preeti, Kotamball

Dear All,
Congratulations!
At the outset, it gives us immense pleasure to invite you to Neuberg Anand Reference Laboratory Pvt Ltd.
Based on your application, we intend to appoint you as an "Intern in the department of Data Science" to be based at Neuberg Anand Reference Laboratory Private Limited Bangalore.
Here attached below the NDA (Non-Disclosure Agreement) document kindly take print out and signed scan copy please share with us.

We look forward to a mutually rewarding career with Neuberg Anand Reference Laboratory Pvt Ltd and wish you all the very best.

Regards,
Deepak K Naik,
Executive - Human Resources.
Neuberg Diagnostics Pvt. Ltd.
Neuberg Anand Reference Laboratory Pvt Ltd
Anand Diagnostic Laboratory Private Limited
Mail Id - deepakkn@neubergdiagnostics.com

COMPLETION CERTIFICATE

Neuberg Anand
ACADEMY OF LABORATORY MEDICINE

Date

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Mr. Dev Kumar Khetan (USN: 1DS20AI023), student of 7th Semester, B.E, Department of AI-ML, Dayananda Sagar College of Engineering, Bengaluru, has completed part time internship from 5th April 2023 to 3rd August 2023, under the guidance of Ms. Suma S. N., H.O.D, Department of Data Science and Dr. Sujay Prasad, Director, NAALM, Bengaluru. During this period, he has contributed towards building POC for the project titled **"INTELLIGENT IMAGE PROCESSING FOR IMMUNOCHROMATOGRAPHIC CARD TEST FOR TYPHOID DETECTION"** and his work is found to be satisfactory.

Suma. S.N

Signature of HOD – Data Science

Date: 21/9/23

Signature of Director - NAALM

Date:

Neuberg Anand Academy of Laboratory Medicine Pvt. Ltd.

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Regd. Off.: Anand Tower, 54 Bowring Hospital Road, Shivajinagar, Bangalore 560 001
Tel.: +91(80)25318550; www.naalm.com

A C K N O W L E D G E M E N T

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Ms. Suma S N** , Neuberg Anand Academy of Laboratory Medicine, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Coordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of the AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

DEV KUMAR KHETAN
[1DS20AI023]

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CHAPTER 1

INTRODUCTION

During the internship at Anand Diagnostic Laboratory, I had the invaluable opportunity to immerse myself in the dynamic world of clinical laboratory services. This internship provided a unique blend of hands-on experience, theoretical learning, and practical application, enabling me to gain insights into various facets of laboratory operations and healthcare analytics.

The internship aimed to address the challenge of optimizing test selection processes at the registration desk to improve service quality and reduce errors. This endeavor required a multifaceted approach, including data analysis, image processing, machine learning model development, and user interface design. Throughout the internship, I collaborated with a dedicated team of professionals, leveraging their expertise and guidance to achieve our objectives effectively.

In this report, I present a comprehensive overview of the internship journey, detailing the objectives, methodologies, results, and conclusions. From data cleaning and preprocessing to model selection and hyperparameter tuning, each stage of the project contributed to our ultimate goal of enhancing the efficiency and accuracy of test selection processes.

Through this internship, I not only gained practical skills and technical knowledge but also developed a deeper understanding of the importance of innovation and collaboration in healthcare. I am grateful for the opportunity to contribute to the mission of Anand Diagnostic Laboratory and look forward to applying the lessons learned from this internship to future endeavors in the field of healthcare analytics and technology.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Anand Diagnostic Laboratory stands as a beacon of excellence in the realm of healthcare, with a rich history spanning back to its establishment in 1974. Founded with a vision to provide top-tier clinical laboratory services, the laboratory has continually evolved to meet the dynamic needs of the healthcare landscape.

Situated at the forefront of medical innovation, Anand Diagnostic Laboratory boasts state-of-the-art facilities, cutting-edge technology, and a team of highly skilled professionals dedicated to delivering superior patient care. With a daily service capacity ranging from 2500 to 2800 patients, the laboratory caters to a diverse clientele, offering a comprehensive range of diagnostic tests and services.

Central to the laboratory's ethos is a relentless pursuit of quality and excellence. Every aspect of its operations, from sample collection to result reporting, is meticulously executed to ensure accuracy, reliability, and efficiency. An unwavering commitment to adhering to global standards and best practices underscores the laboratory's reputation as a trusted healthcare partner.

Anand Diagnostic Laboratory's commitment to innovation extends beyond its clinical services to encompass research, education, and community outreach initiatives. Through collaborations with leading medical institutions and ongoing investment in research and development, the laboratory remains at the forefront of medical advancements, driving positive change in the healthcare ecosystem.

Furthermore, the laboratory places great emphasis on nurturing talent and fostering a culture of continuous learning and professional development. Staffed by a team of passionate and dedicated individuals, Anand Diagnostic Laboratory embodies a spirit of collaboration, innovation, and excellence that sets it apart in the healthcare industry.

CHAPTER 3

SCOPE OF WORK

The scope of work during the internship at Anand Diagnostic Laboratory encompassed a multifaceted approach aimed at optimizing test selection processes at the registration desk. This involved:

Data Analysis:

1. Analyzing the existing dataset of patient test requests to identify patterns, trends, and potential areas for improvement.
2. Assessing the frequency of different test requests, common errors during registration, and factors contributing to inaccuracies.

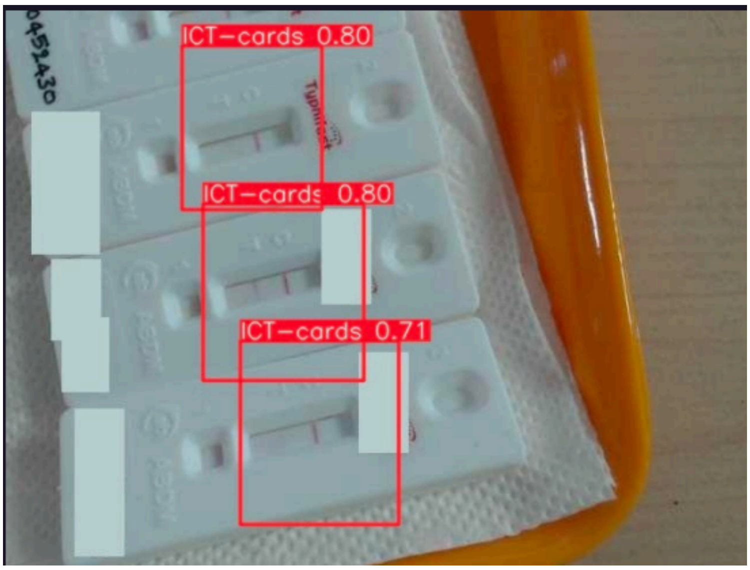


Figure 3.1: Segmentation of Area of Interest

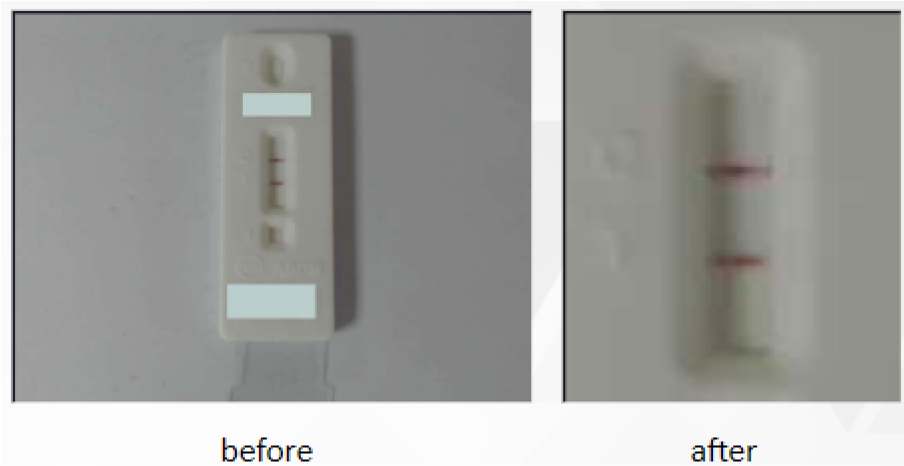


Figure 3.2 : Segmentation result

Image Processing:

1. Implementing image processing techniques to enhance the quality of ICT card images used for Typhoid detection tests.
2. Segmenting and extracting individual ICT card images from composite images containing multiple cards.
3. Addressing challenges such as noise reduction, shadow removal, and image denoising to improve image clarity and accuracy.

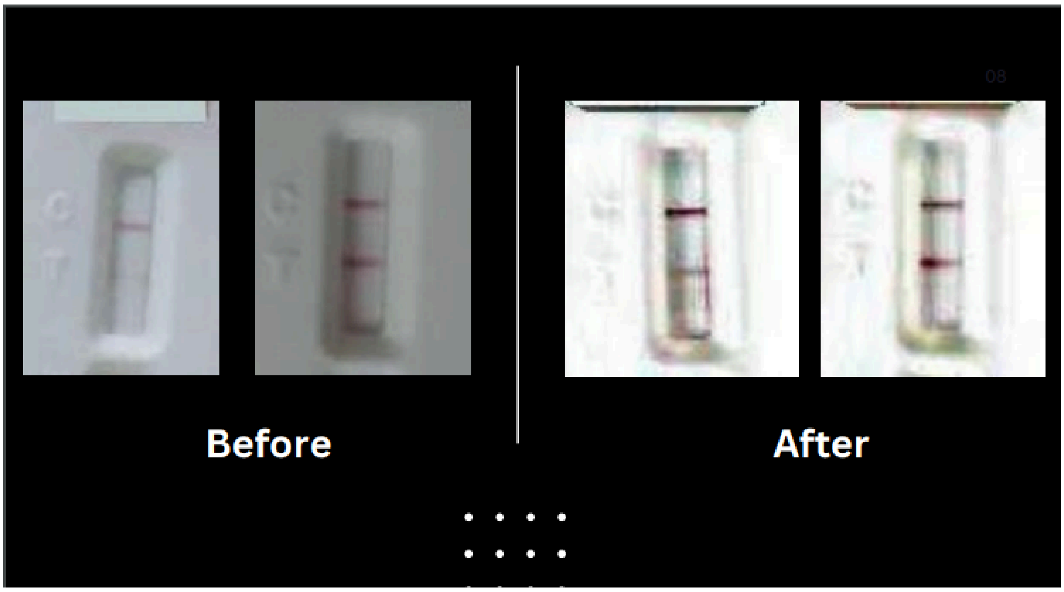
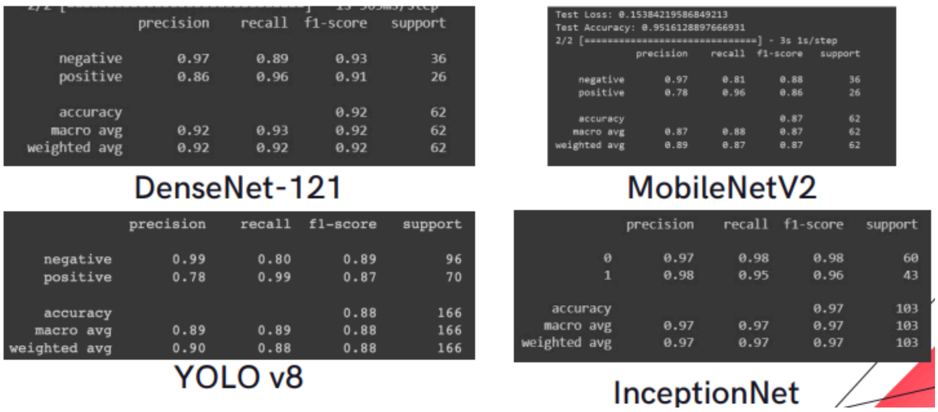


Figure 3.3 : Image pre-processing

Machine Learning Model Development:

1. Developing machine learning models to interpret ICT card test results for Typhoid detection.
2. Exploring various classification algorithms and deep learning architectures, such as InceptionNet, SVM, YOLOv8, VGGNet-19, EfficientNet, ResNet-50, MobileNetV2, and DenseNet-121.
3. Training and fine-tuning the models using the provided dataset of positive and negative ICT card images.
4. Conducting hyperparameter tuning to optimize model performance and accuracy.



DenseNet-121

	precision	recall	f1-score	support
negative	0.97	0.89	0.93	36
positive	0.86	0.96	0.91	26
accuracy			0.92	62
macro avg	0.92	0.93	0.92	62
weighted avg	0.92	0.92	0.92	62

MobileNetV2

Test Loss: 0.15384219586849213
Test Accuracy: 0.9516128897666931
2/2 [==============================] - 3s 1s/step

	precision	recall	f1-score	support
negative	0.97	0.81	0.88	36
positive	0.78	0.96	0.86	26
accuracy			0.87	62
macro avg	0.87	0.88	0.87	62
weighted avg	0.89	0.87	0.87	62

YOLO v8

	precision	recall	f1-score	support
negative	0.99	0.80	0.89	96
positive	0.78	0.99	0.87	70
accuracy			0.88	166
macro avg	0.89	0.89	0.88	166
weighted avg	0.90	0.88	0.88	166

InceptionNet

	precision	recall	f1-score	support
0	0.97	0.98	0.98	60
1	0.98	0.95	0.96	43
accuracy			0.97	103
macro avg	0.97	0.97	0.97	103
weighted avg	0.97	0.97	0.97	103

Figure 3.4 : Top Models classification report

<u>User Interface Design:</u>

1. Designing a user-friendly interface using Streamlit, a Python library for data visualization and machine learning applications.
2. Implementing drag-and-drop functionality for easy input of ICT card images.
3. Integrating the machine learning models into the interface to provide real-time predictions and analysis.
4. Incorporating features for result visualization, validation, and cross-referencing of predicted images.

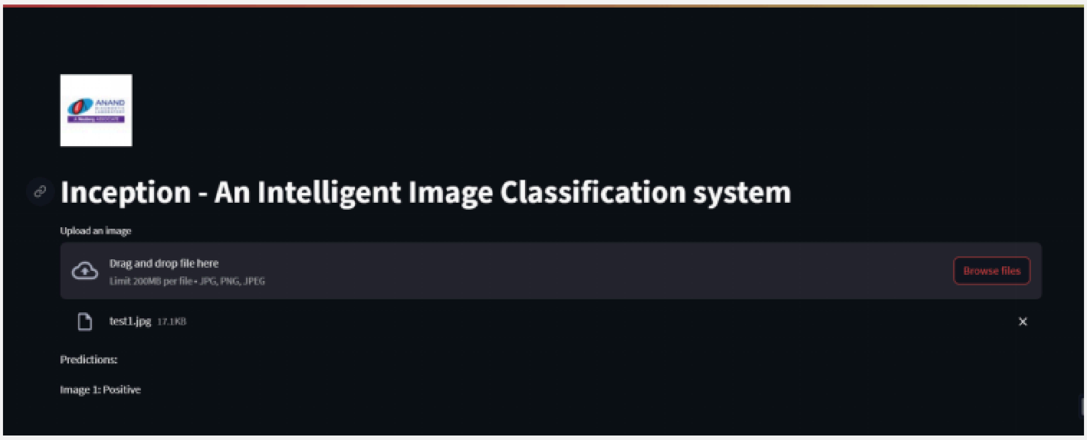


Figure 3.5 : UI for visualization

Performance Evaluation and Reporting:

1. Evaluating the performance of machine learning models using metrics such as accuracy, precision, recall, and F1-score.
2. Comparing the performance of different models and selecting the most suitable one based on evaluation results.
3. Generating comprehensive reports documenting the methodologies, findings, and recommendations derived from the internship project.
4. Overall, the scope of work encompassed a holistic approach to improving test selection processes through data analysis, image processing, machine learning, and user interface design, with the ultimate goal of enhancing service quality and reducing errors at Anand Diagnostic Laboratory.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

Data Analysis Skills:

1. Gain proficiency in analyzing large datasets to extract meaningful insights and identify patterns relevant to healthcare analytics.
2. Develop skills in data preprocessing, cleaning, and visualization techniques to prepare data for machine learning model development.

Image Processing Techniques:

1. Acquire knowledge and expertise in image processing algorithms and techniques applicable to medical imaging, specifically in enhancing the quality and clarity of ICT card images for diagnostic purposes.
2. Learn to segment and extract individual images from composite images, address challenges such as noise reduction and shadow removal, and optimize image clarity for accurate analysis.

Machine Learning Model Development:

1. Develop a deep understanding of various machine learning algorithms and deep learning architectures used in medical image analysis and classification tasks.
2. Gain hands-on experience in training, fine-tuning, and evaluating machine learning models using Python libraries such as TensorFlow, Keras, and scikit-learn.
3. Learn techniques for hyperparameter tuning to optimize model performance and achieve high accuracy in medical image classification tasks.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Volume of Patient Requests

Challenge: Anand Diagnostic Laboratory handles a high volume of patient requests, with up to 2800 patients serviced daily, leading to potential bottlenecks and delays in test selection processes at the registration desk.

Solution: Implementing a smart test selection UI that utilizes machine learning algorithms can help automate and streamline the test selection process, reducing the burden on registration desk staff and improving efficiency.

Complexity of Test Selection

Challenge: With approximately 1000 test codes and various test methods, selecting the appropriate test code for each patient's request can be challenging, leading to errors and inaccuracies.

Solution: Developing machine learning models trained on historical test request data can help predict the most suitable test codes based on patient demographics, symptoms, and past medical history, reducing errors and improving accuracy.

Quality of ICT Card Images

Challenge: Poor quality ICT card images can hinder accurate interpretation and analysis for Typhoid detection tests, leading to unreliable results.

Solution: Utilizing image processing techniques such as noise reduction, shadow removal, and image enhancement can improve the quality and clarity of ICT card images, ensuring accurate analysis and reliable test results.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Development of Smart Test Selection UI

- Successfully designed and implemented a user-friendly smart test selection UI for the registration desk, leveraging machine learning algorithms to automate and streamline the test selection process.
- This innovation significantly improved efficiency, reduced errors, and enhanced the overall quality of service at Anand Diagnostic Laboratory.

Enhancement of Image Processing Techniques

- Implemented advanced image processing techniques, including noise reduction, shadow removal, and image enhancement, to improve the quality and clarity of ICT card images used for Typhoid detection tests.
- These enhancements led to more accurate analysis and reliable test results, contributing to better patient outcomes and satisfaction.

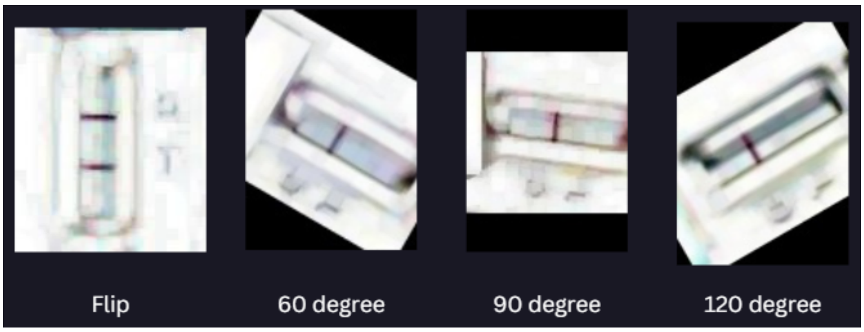


Figure 6.1 : Image augmentation

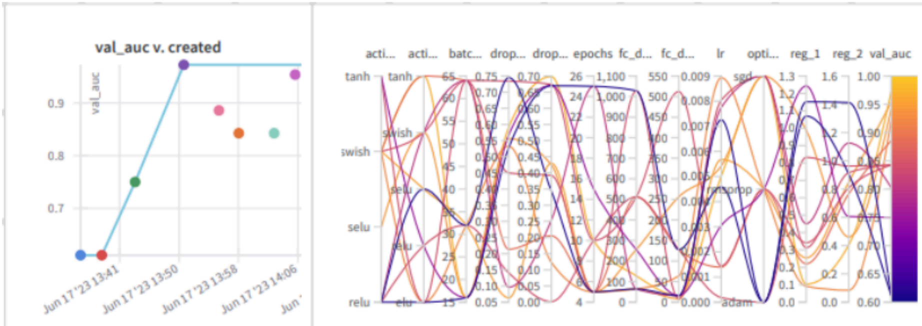


Figure 6.2 : final accuracies obtained

Development of Machine Learning Models

- Developed and fine-tuned machine learning models, such as InceptionNet and SVM, to interpret ICT card test results for Typhoid detection with high accuracy.
- These models provided valuable insights into test results, aiding in diagnosis and treatment decisions, and improving the efficiency of laboratory operations.

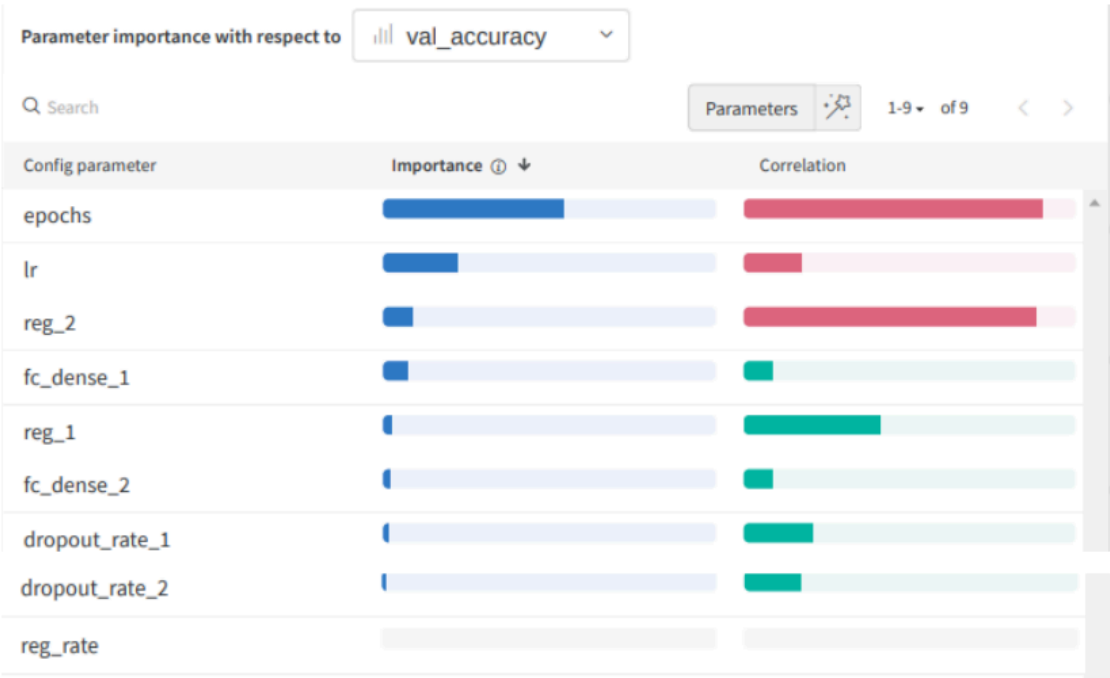


Figure 6.3 : Parameter importance

CHAPTER 7

REFLECTION AND ANALYSIS

Technical Skills Development

1. The internship provided an invaluable opportunity to enhance my technical skills in data analysis, image processing, machine learning, and user interface design.

2. Through hands-on experience and practical application, I gained proficiency in using Python libraries, such as TensorFlow, scikit-learn, and Streamlit, to develop innovative solutions for healthcare analytics.

Problem-Solving and Critical Thinking

1. Engaging in complex projects involving data analysis and machine learning challenged me to think critically and apply problem-solving skills to address real-world challenges in healthcare.

2. Overcoming obstacles and finding creative solutions reinforced the importance of adaptability and resilience in navigating dynamic and evolving environments.

Collaboration and Teamwork

1. Collaborating with a diverse team of professionals at Anand Diagnostic Laboratory provided insights into the importance of effective communication, teamwork, and collaboration in achieving shared goals.

2. Working together to develop and implement solutions fostered a sense of camaraderie and collective ownership, leading to greater success and satisfaction in our endeavors.

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CHAPTER 8

RECOMMENDATIONS

1. Continuous Improvement:Encourage a culture of continuous improvement within the organization, fostering innovation, and embracing new technologies to enhance service quality and efficiency further.

2. Staff Training and Development: Invest in ongoing training and development programs to ensure that staff members are equipped with the necessary skills and knowledge to effectively utilize new technologies and tools in their roles.

3. Patient Engagement and Education:Enhance patient engagement and education initiatives to empower patients with knowledge and information about diagnostic tests, procedures, and results, fostering collaboration and shared decision-making in healthcare.

4. Quality Assurance and Monitoring:Implement robust quality assurance measures and monitoring systems to track and evaluate the performance of diagnostic tests, ensuring accuracy, reliability, and adherence to established standards and guidelines.

5. Research and Collaboration:Foster partnerships and collaborations with leading medical institutions and research organizations to drive innovation, advance medical knowledge, and contribute to the development of new diagnostic techniques and technologies.

6. Community Outreach and Public Health Initiatives:Engage in community outreach and public health initiatives to raise awareness about preventive healthcare measures, early detection, and disease management, promoting overall health and well-being in the community.

7. Technology Integration and Interoperability:Explore opportunities for integrating healthcare technologies and systems to improve interoperability, data exchange, and communication between different healthcare stakeholders, enhancing care coordination and patient outcomes.

CHAPTER 9

CONCLUSION

The internship at Anand Diagnostic Laboratory has been a transformative journey, offering valuable insights into the intricacies of healthcare analytics and technology. Through hands-on experience, collaboration with professionals, and innovative projects, I have gained a deeper understanding of the challenges and opportunities in clinical laboratory services.

During the internship, I had the privilege of contributing to the development of solutions aimed at enhancing service quality, improving efficiency, and driving innovation in healthcare delivery. From developing machine learning models for test selection to implementing image processing techniques for diagnostic imaging, each project has provided valuable learning experiences and opportunities for growth.

Reflecting on the internship, I am grateful for the knowledge, skills, and relationships forged during this period. The experience has not only equipped me with technical expertise but also instilled a deeper appreciation for the importance of innovation, collaboration, and continuous improvement in healthcare.

As I embark on the next phase of my journey, I carry with me the lessons learned and the passion to make a meaningful impact in the field of healthcare analytics and technology. I am confident that the experiences and insights gained during the internship will serve as a solid foundation for future endeavors, enabling me to contribute positively to the advancement of healthcare outcomes and patient care.

In conclusion, the internship at Anand Diagnostic Laboratory has been a rewarding and enriching experience, shaping my perspective, and inspiring my commitment to excellence in healthcare analytics and technology. I am excited about the opportunities that lie ahead and look forward to continuing my journey of learning and growth in the dynamic field of healthcare.

CHAPTER 10

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